

9.2.2

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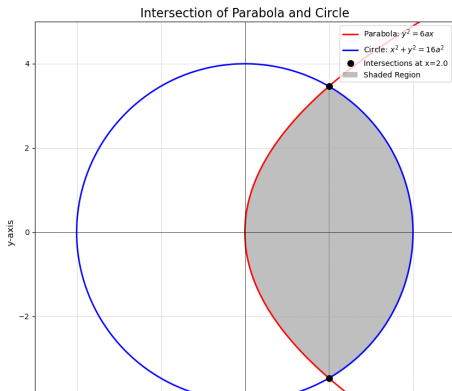
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Question

Find the points of intersection of the following parabola and circle and plot the region:

$$y^2 \leq 6ax \quad (0.1)$$

$$x^2 + y^2 \leq 16a^2 \quad (0.2)$$



Theoretical solution

We express the boundaries in the standard quadratic form $g(x, y) = 0$:

$$y^2 - 6ax = 0 \quad (0.3)$$

$$x^2 + y^2 - 16a^2 = 0 \quad (0.4)$$

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In homogeneous coordinates, using the form $\mathbf{x}^\top C \mathbf{x} = 0$, where $\mathbf{x} = (x \ y \ 1)^\top$, the conic matrices are:

$$C_1 = \begin{pmatrix} 0 & 0 & -3a \\ 0 & 1 & 0 \\ -3a & 0 & 0 \end{pmatrix} \quad (0.5)$$

$$C_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -16a^2 \end{pmatrix} \quad (0.6)$$

Theoretical solution

The intersection points lie on a degenerate conic $C(\mu) = C_1 + \mu C_2$. We find μ by solving $\det(C(\mu)) = 0$.

$$\begin{aligned}\det \begin{pmatrix} \mu & 0 & -3a \\ 0 & 1 + \mu & 0 \\ -3a & 0 & -16a^2\mu \end{pmatrix} &= 0 \\ \implies (\mu)(1 + \mu)\left(-16a^2\mu - \frac{9a^2}{\mu}\right) &= 0 \\ -a^2(1 + \mu)(16\mu^2 + 9) &= 0\end{aligned}\tag{0.7}$$

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The only real root is $\mu = -1$.

Theoretical solution

The equation is found by subtracting (0.4) from (0.3):

$$\begin{aligned}(y^2 - 6ax) - (x^2 + y^2 - 16a^2) &= 0 \\ \implies -x^2 - 6ax + 16a^2 &= 0\end{aligned}\tag{0.8}$$

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Using the general conic matrix representation

$(x \ y) Q \begin{pmatrix} x \\ y \end{pmatrix} + L^\top \begin{pmatrix} x \\ y \end{pmatrix} + C = 0$, we have:

$$Q = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} -6a \\ 0 \end{pmatrix}, \quad C = 16a^2.\tag{0.9}$$

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The full matrix form is:

$$\boxed{(x \ y) \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + (-6a \ 0) \begin{pmatrix} x \\ y \end{pmatrix} + 16a^2 = 0.}\tag{0.10}$$

Theoretical solution

Multiplying by -1 gives $x^2 + 6ax - 16a^2 = 0$, which factors into two linear equations:

$$(x + 8a)(x - 2a) = 0 \quad (0.11)$$

This represents two vertical lines: $x + 8a = 0$ and $x - 2a = 0$.

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$$\boxed{(1 \quad 0) \begin{pmatrix} x \\ y \end{pmatrix} + 8a = 0 \quad \text{and} \quad (1 \quad 0) \begin{pmatrix} x \\ y \end{pmatrix} - 2a = 0.} \quad (0.12)$$

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The line $x = -8a$ does not intersect the parabola $y^2 = 6ax$ in real points (for $a > 0$). We only consider the line $\mathbf{x} = \mathbf{2a}$.

Theoretical solution

We find the intersection of the line $x - 2a = 0$ with the parabola P_1 using the formula $\mathbf{x}_j = \mathbf{h} + \kappa_j \mathbf{m}$.

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The parameters for the line and conic are:

$$\mathbf{h} = \begin{pmatrix} 2a \\ 0 \end{pmatrix}, \mathbf{m} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u} = \begin{pmatrix} -3a \\ 0 \end{pmatrix}, f = 0.$$

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The components for the parameter κ_i are:

$$\mathbf{m}^\top \mathbf{V} \mathbf{m} = 1 \tag{0.13}$$

$$g(\mathbf{h}) = -12a^2 \tag{0.14}$$

$$\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) = 0 \tag{0.15}$$

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Substituting these values gives:

$$\begin{aligned} \kappa_i &= \frac{1}{1} \left(-(0) \pm \sqrt{(0)^2 - (-12a^2)(1)} \right) = \pm \sqrt{12a^2} \\ &= \pm 2\sqrt{3}a \end{aligned} \quad (0.16)$$

Theoretical solution

Finally, we find the intersection points using $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$:

$$\mathbf{x}_1 = \begin{pmatrix} 2a \\ 0 \end{pmatrix} + 2\sqrt{3}a \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2a \\ 2\sqrt{3}a \end{pmatrix} \quad (0.17)$$

$$\mathbf{x}_2 = \begin{pmatrix} 2a \\ 0 \end{pmatrix} - 2\sqrt{3}a \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2a \\ -2\sqrt{3}a \end{pmatrix} \quad (0.18)$$

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The intersection points are:

$(2a, 2\sqrt{3}a)$ **and** $(2a, -2\sqrt{3}a)$

